Shifting Academic Trajectories: How the COVID-19 Pandemic Reshaped Higher Education Choices Across U.S. Regions

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Abstract

The COVID-19 pandemic dramatically disrupted higher education in the United States, revealing stark regional and demographic disparities in academic decision-making. This study investigates how these disruptions influenced higher education plans, focusing on variables such as age, race, gender, household size, income, and regional context. Using data from the Household Pulse Survey (August–September 2020) and Bureau of Economic Analysis (BEA) regional categories, the analysis identifies significant predictors of changes in academic plans. Middle-income households and larger family sizes positively correlated with changes, while older age, white racial identity, and male gender were associated with fewer adjustments. Regional differences emerged as pivotal, with respondents from the Far West and Southwest more likely to alter plans compared to those in the Southeast, reflecting variations in economic conditions and policy responses. Despite modest explanatory power, the findings emphasize the role of demographic and regional factors during systemic disruptions. Key limitations include potential endogeneity, omitted variable bias (e.g., internet access or local policies), and reliance on self-reported data. This research underscores the need for tailored financial aid for middle-income students, efforts to address racial equity, and regionally adaptive educational policies. By highlighting these disparities, the study contributes to understanding how large-scale disruptions reshape educational aspirations and offers actionable insights for policymakers aiming to build resilience and equity in higher education systems.

Introduction

Amid the COVID-19 pandemic, as classrooms shifted online and economic uncertainty gripped households, a new divide emerged,not just between those who could and could not access education, but between regions where aspirations for higher education persisted and those where they faltered. This paper explores how the COVID-19 pandemic, with its far-reaching health and economic consequences, intersected with geographical

disparities to influence America's appetite for higher education. Informally, it investigates how these factors reshaped educational aspirations across different regions. Formally, it asks: How did regional disparities during the pandemic affect individuals' choices regarding higher education attainment?

Being a high school senior in 2020 presented numerous challenges in navigating university choices and priorities. The most significant and enduring hurdle—one that remains relevant to me today as a university senior—was the transition to online courses. Aguilera-Hermida (2020) explores college students' perceptions of adaptation, use, and acceptance of online learning, a factor that I believe could account for significant variance in my study. Their findings indicate that students generally preferred face-to-face learning over online learning, with a moderate correlation between this preference and difficulties in adapting to online platforms. While online learning represents a major technological breakthrough, its acceptance and appeal to prospective students may present significant challenges.

Since this study examines regional variations in impact, I believe Molock and Parchem (2022) offers valuable insights that can enhance our regional models and support the inclusion of race as a control variable in the first model. Their research focused on ethnically diverse college students participating in a STEM career preparation program, analyzing their educational and professional statuses as well as mental health during the COVID-19 pandemic. The findings indicate that the pandemic most significantly affected participants' financial and educational statuses. Additionally, many students described the transition to online learning as "stressful," noting its adverse effects on their academic performance and, for some, their educational plans.

Similar to race, income—specifically middle-income levels ($50,000–$99,000)—was included as a control variable in my initial model. Like the race variable, income levels appear to fluctuate significantly across the regional variables presented in models two and three. Dynarski et al. (2021) supports this observation, highlighting substantial disparities in college admissions and application rates between low-income, high-performing students and their higher-income counterparts with similar academic performance. Although these "low-income" students fall slightly below the middle-income category used in my model, they exemplify the influence of uncertainty, present bias, and loss aversion on college decision-making. By implementing a scholarship program targeting these low-income, high-achieving students—offering an early commitment of free tuition—the study observed significant increases in both application and enrollment rates.

This paper builds on and expands the studies mentioned above, drawing inspiration from their findings while exploring new dimensions of the relationship between demographic and explanatory variables and regional disparities. While prior research, such as Molock and Parchem (2022) and Dynarski et al. (2021), has provided valuable insights into the roles of race, income, and educational challenges during the COVID-19 pandemic, these studies often focus on specific populations or individual-level factors. My work seeks to bridge this gap by examining how these demographic and explanatory variables interact within a regional

context, potentially offering new explanations for observed variances across geographic areas.

This study analyzes changes in higher education plans during the COVID-19 pandemic using data from the Household Pulse Survey. Focusing on responses from August to September 2020, it examines the impact of demographic, socioeconomic, and regional factors on academic decisions. Key variables include age, race, gender, income, and household size, with regional variations assessed using Bureau of Economic Analysis (BEA) categories. Through regression analysis, the study uncovers significant predictors, such as age, income, and household size, while highlighting regional and policy-driven disparities. The findings offer insights into shaping equitable and adaptable educational policies during times of disruption.

Literature Review

This paper is inspired by Dougherty et al. (2022), specifically the idea that the COVID-19 Pandemic had effects on high school students' higher education choices. SLDS data that contains, but it not limited to,student registration, demographics, and information about whether a student participated in a CTE (career and technical education) in a given year, whether they have completed three or more courses in the same substantive area of focus, and whether they earned an industry recognized credential or certificate. A graphical trend analysis along with deviations highlighted in the trends that correspond to the likely influence of COVID-19 were used on this data. The results revealed that students participating in CTE paths were effected far less harshly than Non-CTE Students, notably in their graduation rates(Dougherty et al. 2022).My paper builds on this idea by exploring regional variations in how the COVID-19 pandemic influenced students' higher education choices.

There are plenty of reasons why someone planning to attain higher education would change their plans due to the COVID-19 pandemic. Whether it be concerns about health of themselves or loved ones, closing of campus', or change of class structure, in person to online. Hickey, Hebert, and Webb (2021) collects responses from students in regards to these concerns. Overall, students presented a high level of concern to all questions, with the majority lying in the concerns of themselves of family members contracting COVID-19. The majority of students did not express thoughts of a change in career path, but around a fifth of them did, and around three percent expressed "serious" reconsideration (Hickey, Hebert, and Webb 2021). My study focuses on this subset of students who reconsidered their higher education choices and seeks to determine whether their reasoning can be used to explain to a broader regional population variance.

A concern mentioned by Hickey, Hebert, and Webb (2021) was the fear of contracting the COVID-19 virus, and was the second highest modern concern behind a resurgence regarding the COVID-19 pandemic. Whaley (2022) studies the effect of school visits on transmission of COVID-19. They studied this effect by

building a triple differences model using data from SafeGraph phone mobility data, medical claims data from Castlight Health, and COVID-19 exposure data collected by the CDC. Whaley (2022) found that number of children in the household, number of school visits, and SIP policy were all significant at the one percent level. Unfortunately, Whaley (2022) findings were conducted on lower education. I believe my findings would be similar if not greater for higher education. With the amount of factors that one must consider before making a choice on higher education,I believe that factors such as COVID-19 transmission rates, among others, may have a more pronounced impact when focusing specifically on higher education and examining their effects within individual regions.

A serious concern expressed by students in regards to their choice on higher education is online learning. While offering greater flexibility and access to a broader range of programs,it also presents concerns such as limited face-to-face interaction, feelings level of isolation, and varying levels of engagement. Aguilera-Hermida (2020) explores college students' perception of adaptation, use, and acceptance of online learning. A thirty-six question survey,consisting of qualitative and quantitative questions, of two hundred and seventy students showed that students preferred face-to-face learning relative to online learning and a moderate correlation between preference of face-to-face learning and struggle to adapt to online learning. The model also showed that students were statistically significantly more motivated and had a higher perception of their own skills when face-to-face learning relative to online learning. Online learning is a technological breakthrough, but when it comes to students' learning online learning due to closed campuses have a significant effect on students' perceptions of learning. I believe these students contribute significantly to the regional variance observed in my study, and I encourage others to further explore this concept in future research.

The COVID-19 pandemic has had a seismic effect on higher level education students leading to increased mental health challenges. Son et al. (2020) aims to dive into these challenges by surveying a group of college students from Texas approximately a month after the stay-at-home order in 2020. First, a PSS score was calculated for each student to evaluate their overall stress and anxiety level. Second, the were subjected to a twelve question survey about academics, health, and lifestyle that was examined by using thematic analysis. Findings included: a higher percentage of students concern about academic performance due due to the COVID-19 pandemic, most citing that transition to online learning caused the most concern on academic performance. More significant findings were that: more than half the students expressed concerns about their financial standing being impacted, and many noted that the COVID-19 pandemic has effected or is likely to impact their own current and future employment opportunities (Son et al. 2020). Following Son et al. (2020), I believe potential students could still be effected by these concerns if deciding to pursue higher education. These factors could have significant effects on potential students choice of higher education institution or choice to pursue higher education at all.

As we have seen in Son et al. (2020), mental health was severely effected by the COVID-19 pandemic, this effect is magnified when Molock and Parchem (2022) surveys the effect of COVID-19 on students of color. The survey collected responses from ethnically diverse college students participating in a program that prepared them for STEM careers. Sixteen question surveys were conducted online and asked participants daily well-being, mental health, and experiences of racial discrimination. Interrater reliability was applied to the survey responses. Results from the survey revealed: the COVID-19 Pandemic had the greatest effect on participants' financial and education status, a third of respondents reported a personal job loss, and noted the move to online learning was "stressful" and effected their educational performance and even changed academic plans (Molock and Parchem 2022). Unfortunately, Molock and Parchem (2022) does not expand on respondents' decision to change academic plans, however I believe the significant effect on communities of color, imposed by COVID-19, could be used to explain regional differences; one of the main topics of this study.

While the COVID-19 pandemic has changed the landscape of higher education, it is critical that we understand and expand on other factors play role in higher education choices. Dynarski et al. (2021) studies a large disparity in admissions, and even applications, among low-income high performance students relative to their higher income similar performance counterparts. High-achieving and low-income students in Michigan are located using longitudinal- student level data including all public high schools in the state. Students were considered "high-achieving" when they met the threshold of a 3.3 GPA and 1100 SAT score (scores indirectly acquired for admission to highly selective colleges) A student deemed as low income had to qualify for a free or reduced price lunch ($32,630 to $46,435 annual house hold income). When low-income and high-achieving students are identified they qualify for the HAIL scholarship (early commitment of free tuition). The effect of HAIL on application, admission, enrollment, and persistence at the University of Michigan by comparing the outcomes of treatment and control students, estimating the models by OLS (Dynarski et al. 2021). Results of the HAIL scholarship substantially increased application (68% vs 26%) and enrollment (27% vs 12%) rates. These results show that uncertainty, present bias, and loss aversion affect student's college decisions that could possibly be responsible for unwanted variance in my model.

In today's higher education landscape, many colleges highly cater to students' preferences for consumption such as amenities often at the expense of academic quality. Jacob, McCall, and Stange (2018) investigates whether demand-side market pressure explains this emphasis on amenities relative to quality. Student level data form two nationally representative cohorts of high school seniors is combined with college level data on nearly all 4-year colleges in the United States. Student data was collected from 2 national Longitudinal studies, one from 1992 and the other from 2004. College level data regarding "quality" was expenditures on instruction and academic support per full-time equivalent (FTE) student from the IPEDS Finance Survey

which includes all expenses for all forms of instruction. Measure of consumption amenities is current spending and auxiliary enterprises also from the IPEDS Finance Survey. Using a discrete choice model, Jacob, McCall, and Stange (2018) finds that most students appear to value consumption amenities while the taste for academic quality is present in higher-achieving students. College allocation of funding could also play a part in students' choices of higher education and cause unexplained variance in my model. However, this data was collected from cohorts in 1992 and 2004, times that have never seen a pandemic on the scale of COVID-19.

Choice of college is a complex problem, with many intended and unintended consequences that come along with it. Many 17-18 year old high school students lack the ability and guidance to make the optimal choice for them and are vulnerable to external pressures. Mulhern (2021) analyzes the introduction of a low-cost personalized college admissions information tool and how that can change student's choice of college. The main sample is made up of around 8,000 students who graduated from a district high school in 2015 or 2016 in a Mid-Atlantic state. The school district provides data on student: demographics, coursework, grades and test scores. Application data is collected from Naviance and admission decisions are self reported in a graduating student survey. Mulhern (2021) estimates the causal impact of access to a scattergram using a regression discontinuity design. Students are significantly more likely to apply to colleges with visible admissions information relative to those that do not, this information being visible made application rates increase by 0.27 percentage points effectively increasing the probability of applying to a college by 20% (Mulhern 2021). I believe the technology examined by Mulhern (2021) may introduce unexplained variance in my model. However, I contend that some of this variance could be accounted for by regional differences in infrastructure and scholarship opportunities.

Similar to perceptions of learning. This unprecedented time had a great impact on enrollment, graduation decisions, and expectations regarding future employment. Aucejo et al. (2020) surveyed 1446 students from Arizona State University and collected responses on non-academic, academic and future outcomes. Using an empirical analysis using treatment effects and fixed effects for withdrawal of a class, delayed graduation, and change of major found statistically significant results. The model yielded economically and statistically significant treatment effects for all outcomes. A negative relationship was found between COVID-19 being present and educational performance outcomes and a statistically significant of students reported a change in graduation plans, withdrawal of class, or change of major. This paper also examines similar outcomes, reinforcing my argument that the COVID-19 pandemic influenced students' educational choices, potentially varying across geographic regions, which I study.

Data

The data for this paper comes from the Household Pulse Survey, conducted by the United States Census Bureau. This survey gathers responses from approximately 30,000 participants on various topics, including vaccination status, household size, and changes in academic plans. For this study, I will focus on survey responses collected between August 19, 2020, and September 26, 2020 (weeks 13-15).

The survey includes information on participants' higher education plans and any changes to these plans, along with the reasons for such changes. My analysis will focus on the variability in changes to higher education plans across different geographic regions provided in the data.

The variables of interest include demographic factors such as age, race, gender, region, and household size. Other variables are more complex, such as Bureau of Economic Analysis (BEA) regions, employment status, and changes to academic plans. The age variable was not originally included in the dataset; therefore, I derived it by subtracting the participant's birth year from 2021. I anticipate a negative relationship between age and changes to academic plans, as older participants are less likely to pursue higher education or may be less responsive to changes in the educational landscape than younger participants.

Race is included as a categorical variable, coded from 1 to 4 to represent different racial categories. I do not expect race to have a significant impact on changes to higher education plans, as I believe that no particular racial group faced disproportionate effects in this area due to COVID-19. Gender is similarly coded, with "Male" represented as 1 and "Female" as 2. I do not expect gender to be a significant predictor in this study, as I believe both genders experienced similar impacts on their higher education choices during the pandemic.

The region variable categorizes responses as follows: 1 for "Northeast," 2 for "South," 3 for "Midwest," and 4 for "West." Household size is also included as a variable. I expect household size to show a significant relationship with changes to academic plans, as family dynamics and responsibilities can influence educational decisions. While I am uncertain which specific regions will show greater significance, I believe a regional correlation exists.

To further analyze geographic impacts, I created BEA region "dummy" variables by assigning each state to one of eight BEA regions. States received a value of 1 if they fell within a specific region and 0 otherwise. This approach is intended to provide a more granular understanding of regional variations beyond the broader region categories.

The employment status variable was derived from the "anywork" variable in the initial dataset, which indicated whether participants had been employed in the past seven days (1 for employed, 0 for unemployed). I excluded participants marked as 0, as those who had not been employed in the last seven days are less

relevant to my study on higher education choices. I do not anticipate employment status to be a significant factor due to the widespread job instability during the pandemic.

The key outcome variable in my analysis is "Change to Academic Plans." This variable was constructed using several components of the initial dataset, particularly the variables "psychng2-7," which record changes to academic plans. I excluded "psychng1," as it corresponded to "no change," which is not pertinent to the study's focus on understanding the factors driving changes in academic plans.

Ideally, the dataset would focus on responses from individuals aged 18-25, a range most likely to include those pursuing higher education. However, narrowing the sample in this way would limit the dataset's representativeness. Additionally, the reasoning behind changes to academic plans could be more easily analyzed if it were divided into separate binary (dummy) variables. While this would facilitate analysis, it would also add complexity to an already extensive dataset containing over 200 variables. Table 1 shows summary statistics.

Table 1: Summary Statistics

Statistic	Mean	St. Dev.	Min	Max	N
Age	51.92	15.89	19	89	318,372
White	0.81	0.39	0	1	318,372
Non-White	0.19	0.39	0	1	318,372
Male	0.41	0.49	0	1	318,372
Employed	0.62	0.49	0	1	318,372
Less than Highschool	0.01	0.08	0	1	318,372
Some Highschool	0.01	0.12	0	1	318,372
GED	0.12	0.32	0	1	318,372
Some College	0.22	0.41	0	1	318,372
Associate's Degree	0.10	0.31	0	1	318,372
Bachelor's Degree	0.29	0.45	0	1	318,372
Graduate Degree	0.25	0.43	0	1	318,372
Far West	0.19	0.39	0	1	318,372
Rocky Mountians	0.09	0.29	0	1	318,372
Southwest	0.11	0.31	0	1	318,372
Plains	0.11	0.31	0	1	318,372
Southeast	0.21	0.40	0	1	318,372
Great Lakes	0.10	0.29	0	1	318,372
Mideast	0.11	0.32	0	1	318,372
New England	0.09	0.29	0	1	318,372
Change in Academic Plans	0.15	0.36	0	1	318,372
Household Size	2.79	1.50	1	10	318,372
Low Income	0.23	0.42	0	1	318,372
Middle Income	0.25	0.43	0	1	318,372
High Income	0.30	0.46	0	1	318,372

As illustrated in Table 1, our respondents represent a diverse demographic. The age of respondents spans a broad range, with the youngest at 19 years and the oldest at 89 years. The standard deviation of

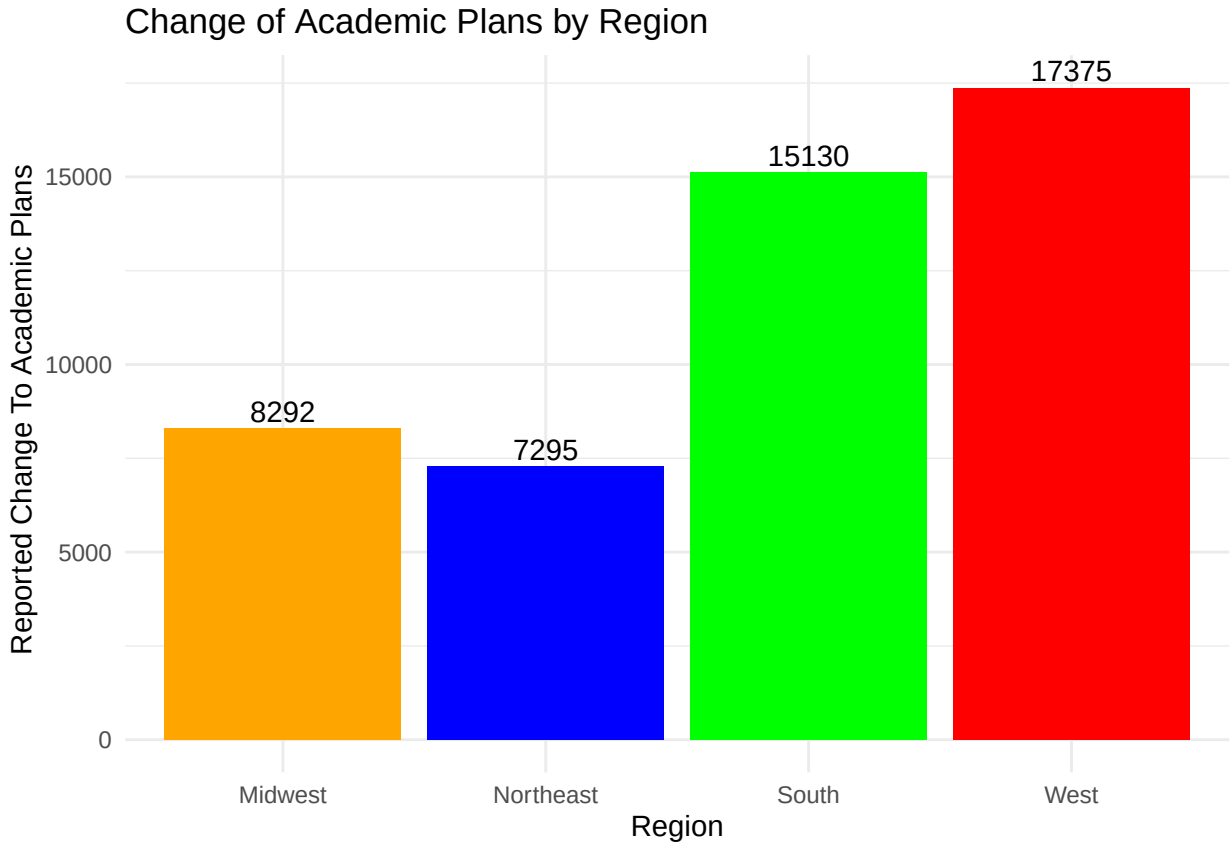
approximately 16 years indicates considerable variation, which is expected given this wide age range. The average age of respondents is around 52 years.

Our gender statistics reveal a slight skew towards female respondents, with an average of 1.59, where 1 represents "Male" and 2 represents "Female." This suggests a modestly higher representation of women in our sample. In contrast, the race variable is more imbalanced, showing a strong skew towards respondents identifying as White, with an average of 1.34. This suggests that other racial groups, represented by values 2 to 4, were less frequently reported.

The "Change in Academic Plans" variable is a binary indicator, reflecting that approximately 15% of respondents reported a change in their academic plans. Household size, similar to age, displayed considerable variability, with a maximum household size of 10 and an average household size of around 3 people.

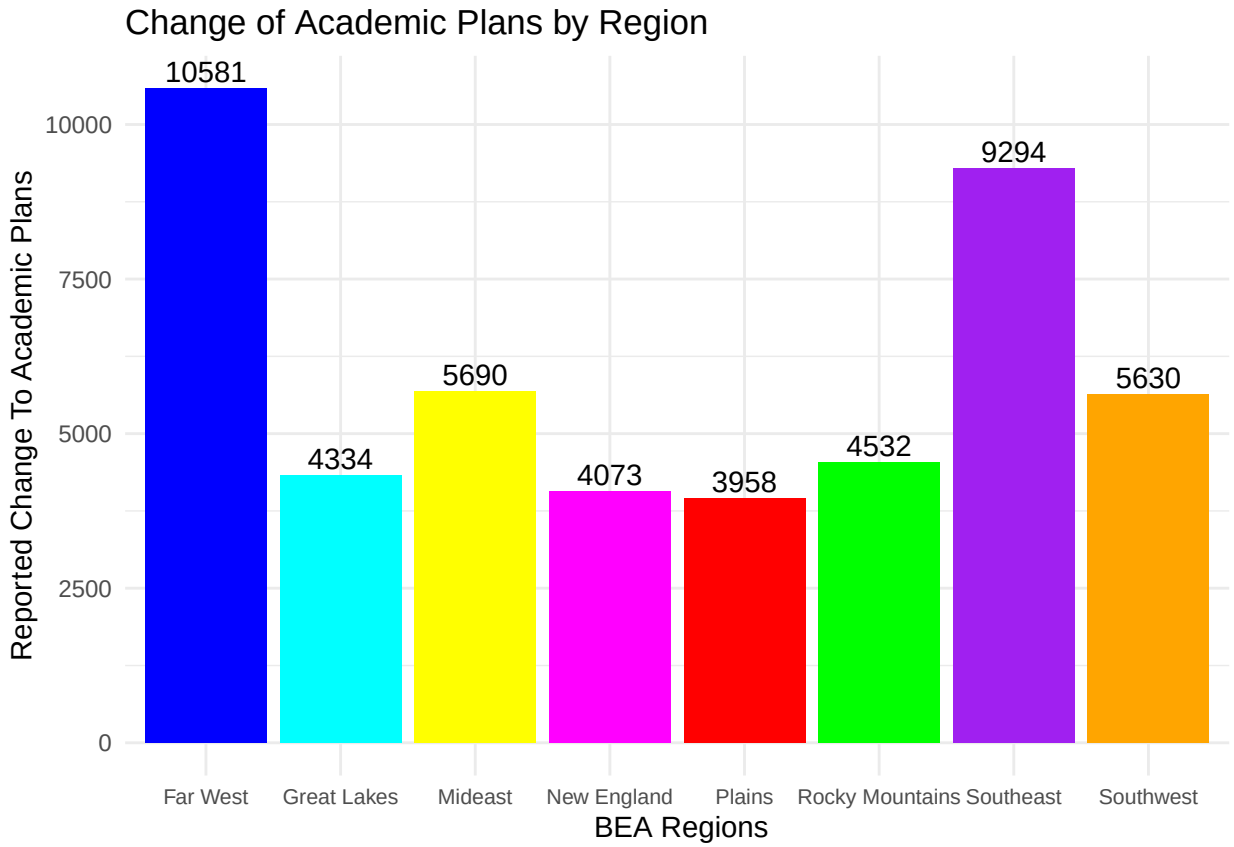
The education level variable is particularly noteworthy, with the average respondent having achieved education levels around an associate degree. The employment variable reflects a representative share of the population, indicating a 62% likelihood of respondents being employed at the time of the survey.

Finally, our region variable shows a clustering of respondents in the Southern and Midwestern regions, with an average of 2.69, suggesting a higher concentration in these areas compared to others.



The West region showed the highest number of students with academic plan changes, while the Northeast

region had the lowest. These results highlight broad regional disparities, potentially linked to factors such as socioeconomic conditions or educational infrastructure. Geography appears to play a significant role in how students adapted their plans during this period. The number of participants differed across regions, with larger regions like the West and South contributing more data compared to smaller regions such as the Northeast and Midwest. While this variation doesn't undermine the observed trends, it highlights the need to interpret results with consideration of these regional differences in sample size.



Breaking regions into subcategories revealed greater variation. For example, Southeast and Far West regions had notably higher counts compared to Plains and New England. These granular differences suggest subregional factors, like local policies or access to resources, significantly influenced outcomes. The detailed view emphasizes the importance of studying regional nuances to fully understand the impact on academic plans. While the graphs reveal clear trends, it's important to note that the number of participants varied across regions. Larger regions, such as Far West and South East, provide more robust insights, whereas smaller regions, like Rocky Mountain and New England, might be more sensitive to outliers. This variance does not diminish the trends observed but should be considered when interpreting the results.

Method

I utilize the following linear probability model:

$$ChangeInAcademicPlans_{h,r,w} = \beta_0 + \beta_1 Region_{h,r,w} + X'_{h,r,w} + \alpha_w + \epsilon \tag{1}$$

The outcome variable, $ChangeInAcademicPlans$, captures respondents' reports of changes to their academic plans during the COVID-19 pandemic.

The variable *Region* is a set of dummy variables I created from the state indicator. I grouped all fifty states (including DC) into their respective BEA regions: Far West, Rocky Mountains, Southwest, Plains, Southeast, Great Lakes, Mideast, and New England. *Region* is my primary variable of interest, as I am interested in the potential impact of regional differences on academic choices. I do not expect a statistically significant relationship across the eight BEA regions, as COVID-19 impacted the entire United States. However, it remains possible that one region was more affected than another.

X' is a vector of control variables that influence the outcome variable, but are not the main variable of interest (*Region*). This vector includes basic demographic variables—such as age, race, and gender—as well as socioeconomic variables, including education level, income, and household size. I chose these controls to account for potential confounding factors that might influence the primary relationship of interest. Demographic factors like age, gender, and race provide insight into group-level differences, while socioeconomic variables like income and education reflect variations in access to resources and opportunities. By including these controls, the analysis gains accuracy and robustness, ensuring that observed effects are not driven by underlying demographic or socioeconomic disparities.

Among the basic demographic variables (*age*, *race*, and *gender*) included in X', I expect to see little to no statistical significance or relationship with the outcome variable. However, for *education*, which measures the highest level of educational attainment (ranging from less than high school to a graduate degree), I anticipate a strong, negative, and significant effect on $ChangeInAcademicPlans$, as individuals with higher educational attainment are likely less inclined to alter their academic plans. Similar to *age*, *education* may exhibit a quadratic trend due to its wide variance. *Household Size* represents the number of individuals in the respondent's household and could positively correlate with $ChangeInAcademicPlans$, as larger households may suggest a higher likelihood of dependents, potentially increasing the flexibility of educational plans.

In my analysis, I include both regional and time fixed effects. Regional fixed effects account for characteristics specific to each region that do not change over time but may impact the outcome variable. These could include cultural norms, regional economic conditions, policy differences, infrastructure, and access to public services. Similarly, time fixed effects capture factors that vary over time but are consistent across all regions

or entities within each period, such as macroeconomic trends (e.g., inflation or recessions), technological advancements, or broad policy changes that affect all regions equally.

I cannot include individual fixed effects because my data source is cross-sectional, not panel data, which would require repeated measures for the same respondents over time.

As robustness checks, I will run the original model and two additional versions with minor adjustments. For the first robustness check, I will reconfigure the regional fixed effects to the standard North, East, South, and West classifications. This adjustment may reveal relationships obscured by the more granular eight-region breakdown. Second, given the high variance in *age*, I will exclude age outliers and rerun the regression to identify any relationships that may have been distorted by extreme values.

To analyze these variables, I will use Ordinary Least Squares (OLS) Regression. OLS is widely understood and straightforward, enhancing result interpretability. Additionally, OLS is versatile, easily accommodating multiple dummy variables as independent factors. However, OLS relies heavily on assumptions, such as normality and the absence of multicollinearity; violations of these assumptions can distort results. Moreover, OLS does not address endogeneity concerns, such as omitted variable bias, which may affect the model's accuracy.

Results

In the first model, I ran a linear regression on my outcome variable, controlling for several demographic and background factors. Most of the variables included in the regression yielded statistically significant coefficients.

The age variable followed expected results, showing a negative and statistically significant relationship with the outcome variable at the 1% level. Specifically, the coefficient was -0.002, meaning that for each additional year of age, the probability of a respondent changing their academic plans decreased by 0.002 percentage points. Age remained robust across all models, with no change in the coefficient or significance.

The "white" variable also demonstrated a negative and statistically significant relationship with the outcome variable at the 1% level, with a coefficient of -0.040. This implies that white respondents were 0.040 percentage points less likely to report changes to their academic plans compared to non-white respondents. Like the age variable, the coefficient for race showed minimal variation (0.001) and maintained its statistical significance across all models, supporting its robustness.

The "male" variable yielded a negative and statistically significant relationship with the outcome at the 1% level, with a coefficient of -0.016. This suggests that male respondents were 0.016 percentage points less likely to change their academic plans. While the coefficient remained robust between the first two models, it

Table 2: Regression Results - Linear Probability Model

	Dependent Variable:		
	Change to Academic Plans		
	(1)	(2)	(3)
Age	-0.002^{***}	-0.002^{***}	
	(0.00004)	(0.00004)	
1-3 Quartile Age			0.001^{***}
			(0.0001)
White	-0.040^{***}	-0.043^{***}	-0.042^{***}
	(0.002)	(0.002)	(0.002)
Male	-0.016^{***}	-0.016^{***}	-0.027^{***}
	(0.001)	(0.001)	(0.002)
Working	-0.007^{***}	-0.008^{***}	-0.002
	(0.001)	(0.001)	(0.002)
Midlevel Income	0.051^{***}	0.050^{***}	0.060^{***}
	(0.001)	(0.001)	(0.002)
Household Size	0.030^{***}	0.030^{***}	0.043^{***}
	(0.0004)	(0.0004)	(0.001)
Bachelor's Degree	-0.003^{**}	-0.003^{**}	0.012^{***}
	(0.001)	(0.001)	(0.002)
Controls	Yes	Yes	Yes
Time Fixed Effects	Yes	Yes	Yes
Observations	318,372	318,372	167,597
R^2	0.043	0.043	0.038

Note: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

1-3 Quartile Age only collects age values in and between first and third quartile, eliminating outliers.

Table 3: Regression Results - Linear Probability Model

	Dependent Variable:
	Change in Academic Plans
northeast	-0.014^{***}
	(0.002)
midwest	-0.036^{***}
	(0.002)
south	-0.017^{***}
	(0.002)
Controls	No
Census Region and Time Fixed Effects	Yes
Observations	318,372
R^2	0.043
Note:	*p<0.1; **p<0.05; ***p<0.01

increased slightly in the third model (0.09 points) while maintaining statistical significance.

The employment variable, which measures whether a respondent had worked in the past seven days, also showed a negative and statistically significant relationship with the outcome at the 1% level. Specifically, employed respondents were 0.007 percentage points less likely to change their academic plans, which aligns with expectations. While the variable remained significant and robust in Models 1 and 2, it lost statistical significance and saw a slight decrease in the coefficient in Model 3, after excluding an outlier in the age variable.

The "midlevel income" variable represents respondents who reported an annual income between $50,000 and $90,000. This variable showed a positive and statistically significant relationship with the outcome at the 1% level, with a coefficient of 0.051. This means respondents within this income bracket were 0.051 percentage points more likely to report changes in their academic plans. Midlevel income remained significant across all models, with a small increase in the coefficient in Model 3 (0.09 points).

The household size variable, which reflects the number of people living in a respondent's household, revealed a positive and statistically significant relationship with the outcome at the 1% level, with a coefficient of 0.030. Larger households may require respondents to consider the needs of dependents, possibly reducing their focus on education. Like midlevel income, household size maintained statistical significance across all three models and saw a small increase in the coefficient in Model 3 (0.013 points).

The "bachelor's degree" variable showed a negative but statistically significant relationship with the outcome at the 10% level, with a coefficient of -0.003. This suggests that respondents with a bachelor's

Table 4: Regression Results - Linear Probability Model

	Dependent Variable:		
	Change in Academic Plans		
	(1)	(2)	(3)
farwest	0.029***		0.030***
	(0.002)		(0.003)
rockymount	0.003		−0.001
	(0.002)		(0.004)
southw	0.023***		0.022***
	(0.002)		(0.003)
plains	−0.026***		−0.024***
	(0.002)		(0.003)
greatlake	0.00001		0.003
	(0.002)		(0.004)
mide	0.013***		0.017***
	(0.002)		(0.003)
neweng	0.007***		0.006
	(0.002)		(0.004)
Controls	No	No	No
BEA Region and Time Fixed Effects	Yes	Yes	Yes
Observations	318,372	318,372	167,597
R^2	0.043	0.043	0.038
Note:		*p<0.1; **p<0.05; ***p<0.01	

degree were slightly less likely to change their academic plans. However, the robustness of this variable was questionable, as its relationship flipped from negative to positive in Model 3, with a coefficient of 0.015, while still maintaining significance at the 10% level.

In Model 2, I introduced regional fixed effects by grouping states into four U.S. census regions: Northeast, Midwest, South, and West (excluded as the reference region). The results were as follows:

The Northeast variable revealed a negative and statistically significant relationship with the outcome at the 1% level, with a coefficient of -0.014. This suggests that respondents from the Northeast were 0.014 percentage points less likely to change their academic plans compared to those from the West. However, the coefficient of the relationship was smaller than for the other census regions.

The Midwest variable exhibited a negative and statistically significant relationship with the outcome at the 1% level, with a coefficient of -0.036. This implies that respondents from the Midwest were 0.036 percentage points less likely to report changes to their academic plans compared to those in the West.

The South variable showed a negative and statistically significant relationship with the outcome at the 1% level, with a coefficient of -0.017. Leading to less likehihood of changes to academic plans from somone in the south compared to the West.

In Model 3, I refined the regional analysis by grouping states into eight different Bureau of Economic Analysis (BEA) regions and replaced the age variable included in model 1 with a refined age variable that drops outliers by collecting values only within the 1st and 3rd quartile. The BEA regions are: Far West, Rocky Mountains, Southwest, Southeast (excluded as the reference region), Plains, Great Lakes, Mideast, and New England. The results are as follows:

The Far West region showed a positive and statistically significant relationship with the outcome at the 1% level, with a coefficient of 0.029. This means that respondents from the Far West were 0.029 percentage points more likely to change their academic plans compared to those from the Southeast. The coefficient remained stable and significant across both models, demonstrating its robustness.

The Rocky Mountains region showed a positive but statistically insignificant relationship with the outcome, with a coefficient of 0.003. The relationship remained weak and flipped from significant to insignificant in Model 3, suggesting a reduced impact.

The Southwest region displayed a positive and statistically significant relationship with the outcome at the 1% level, with a coefficient of 0.023. This suggests that respondents from the Southwest were 0.023 percentage points more likely to change their academic plans. Similar to the Far West, the coefficient remained stable and significant across both models.

The Plains region exhibited a negative and statistically significant relationship with the outcome at the 1% level, with a coefficient of -0.026. This indicates that respondents from the Plains were 0.026 percentage

points less likely to change their academic plans compared to those from the Southeast. The coefficient remained robust across models.

The Great Lakes region showed an extremely small coefficient of 0.00001, indicating an insignificant relationship with the outcome. Despite this, the coefficient of the coefficient increased significantly in Model 3, though it remained statistically insignificant.

The Mideast region displayed a positive and statistically significant relationship with the outcome at the 1% level, with a coefficient of 0.013. Like other BEA regions, the Mideast saw a small increase in the coefficient across models, but retained its significance.

The New England region showed a positive and statistically significant relationship with the outcome at the 1% level, with a coefficient of 0.007. However, despite its initial significance, the New England variable lost all significance in Model 3.

The R^2 values across the models varied between 0.043 and 0.038, indicating that the models explained only a small proportion of the variance in the outcome variable. While the R^2 did not fluctuate significantly across models, the models still accounted for only a modest amount of the total variance.

The results from this analysis, while showing small coefficient sizes, offer important insights into the factors influencing changes in academic plans. Though the R^2 values indicate a modest explanatory power, the statistical significance of the relationships suggests these variables consistently influence educational decisions.

The economic significance of these relationships lies in their potential cumulative impact. For example, while the age variable shows a small coefficient of -0.002, the effect across large populations could lead to noticeable shifts in educational behavior over time. Similarly, the negative relationship between race (white) and academic changes suggests that racial disparities in access to resources and support systems may affect the ability to alter educational plans. Income, particularly midlevel income, has a more economically significant effect, with respondents in this group 0.051 percentage points more likely to change their plans, highlighting the role of financial flexibility in shaping educational trajectories.

Potential explanations for these findings include age-related stability in career and education paths, financial flexibility allowing for academic changes among middle-income individuals, and racial disparities in institutional support. Older respondents may be less likely to change academic plans due to increased responsibilities, while white respondents may have more stable pathways because of greater access to resources. Income appears to enable more flexibility in academic decisions, with middle-income individuals better able to absorb the costs and risks of changing educational goals.

The implications of these results are significant for policy design. For instance, financial aid programs could be structured to allow middle-income students more flexibility to pursue changing academic paths

without the burden of debt. The racial disparities observed suggest a need for policies that address unequal access to support systems, such as mentorship and career development, to ensure all students have equal opportunities to adjust their educational goals. Additionally, policies that support older or non-traditional students—such as flexible scheduling and childcare services—could reduce barriers to academic changes for those with familial or financial responsibilities.

Regionally, the coefficients displayed reasonable stability across the eight BEA regions—Far West, Rocky Mountains, Southwest, Plains, Great Lakes, Mideast, Southeast, and New England. Notably, most coefficients were positive, indicating that, compared to the reference group (Southeast), individuals or institutions in these regions were more likely to experience changes to academic plans. This trend highlights regional differences in responsiveness to educational choice policies and reforms.

The Southeast, serving as the reference group, may exhibit less change due to its unique demographic composition, policy landscape, or cultural attitudes toward traditional education systems. In contrast, regions with positive coefficients, such as the Far West or Mideast, often feature more diverse educational ecosystems, greater policy support for alternative education models, or stronger demand for innovation in schooling options.

Demographic factors play a central role in explaining these patterns. For instance, regions with younger or more transient populations, such as the Southwest or Rocky Mountains, might have a greater propensity for adopting new educational plans. Similarly, areas with higher socio-economic diversity, like the Far West, may display a stronger push for educational flexibility to meet varied needs.

Policy differences further drive these regional trends. States in regions with positive coefficients often have more robust frameworks supporting educational choice, including expanded access to charter schools, voucher programs, and online education. These policy environments encourage experimentation and adaptation, leading to more frequent changes in academic plans.

Cultural influences also shape regional disparities. Regions with a history of valuing educational innovation or individual choice, such as the New England and Great Lakes regions, may foster environments where changes to academic plans are more likely. Conversely, the Southeast may exhibit more resistance to change due to deeply rooted traditions in public education.

In conclusion, while the individual effects are small, the consistency of these results points to important patterns in how demographic and socioeconomic factors influence educational decisions. These findings suggest a need for targeted policy interventions to support greater flexibility and equity in educational systems.

Conclusion

This study provides valuable insights into the factors influencing changes in higher education plans during the COVID-19 pandemic, leveraging data from the Household Pulse Survey conducted by the U.S. Census Bureau. By examining responses from August 19 to September 26, 2020, the analysis highlights the complex interplay of demographic, socioeconomic, and regional factors that shaped educational decisions during this period of disruption. Through the use of regression analysis, significant relationships emerged between changes in academic plans and variables such as age, race, gender, household size, income, and regional categories. Notably, midlevel income and larger household size positively influenced changes in academic plans, while older age, white racial identity, and male gender were associated with a lower likelihood of altering educational trajectories.

Regional analysis, with an emphasis on Bureau of Economic Analysis (BEA) categories, revealed important geographic variations. Respondents from regions such as the Far West and Southwest were more likely to adjust their plans compared to those in the Southeast, reflecting potential differences in regional economic conditions, policy environments, and cultural attitudes. These findings emphasize the importance of regional and demographic contexts in shaping responses to systemic disruptions like the pandemic.

While the study faced limitations, such as modest explanatory power (indicated by low R^2 values) and potential challenges with variable complexity, the robustness of key coefficients underscores the reliability of its findings. The results suggest that demographic and socioeconomic disparities, coupled with regional differences, played critical roles in determining the adaptability of educational plans during the pandemic.

While this study provides valuable insights into how demographic and regional factors shaped higher education plans during the COVID-19 pandemic, several limitations warrant discussion. The analysis may face potential endogeneity issues or omitted variable bias (OVB), as some unobserved factors—such as access to reliable internet, local unemployment rates, or the severity of regional lockdown measures—could influence both the predictors and outcomes. Additionally, self-reported data from the Household Pulse Survey might introduce measurement error, potentially affecting the reliability of the results.

The absence of certain variables in the survey, such as family financial aid strategies, individual access to resources, or local policy responses, limits the ability to fully account for the complex factors driving regional disparities. Finally, the study's focus on responses collected between August and September 2020 captures only a snapshot of a dynamic period, potentially omitting longer-term adjustments to academic plans as the pandemic unfolded.

Addressing these limitations in future research would provide a more comprehensive understanding of how systemic disruptions affect educational choices, enhancing the robustness and applicability of findings.

The implications of these findings are significant for policymakers and educational institutions. Efforts to promote equity in higher education should address racial disparities, provide financial aid tailored to middle-income students, and support flexible educational frameworks for non-traditional learners. Regional policy considerations should account for the unique needs and challenges of different areas, fostering adaptable and inclusive systems that enable students to navigate disruptions more effectively. By understanding the factors that drive changes in academic plans, this research contributes to the broader conversation on building resilience and equity in education during times of crisis.

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